# IR-Ring V1.1

## STM32

USB: max. 0.5A
IR sensors must be disabled when powered via USB

### ESD protection

### Connector for STLINK-V3MINI

Lego-Mindstorms I2C protocol requires 82k pullups

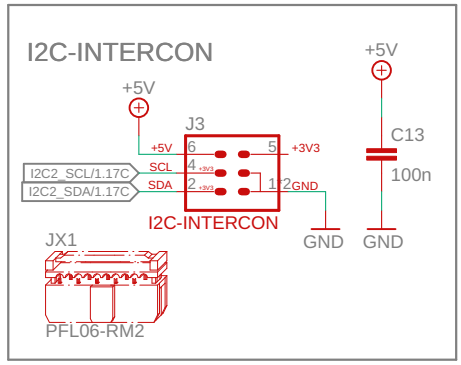
DIP4 ON to enter DFU mode

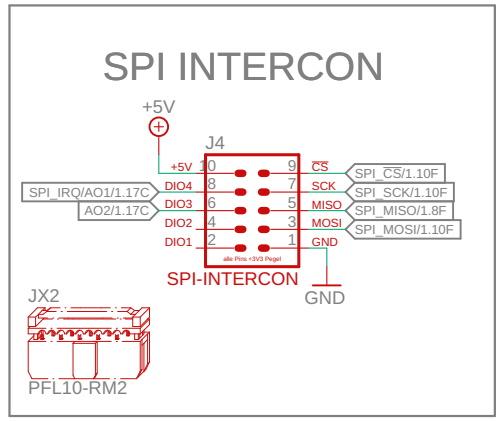
STM32G474MET

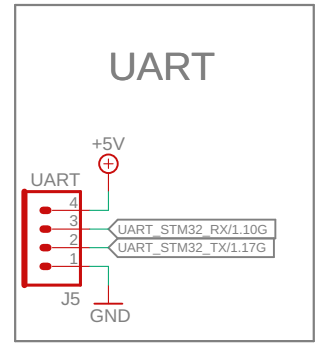
ASDTDV-24.000MHz

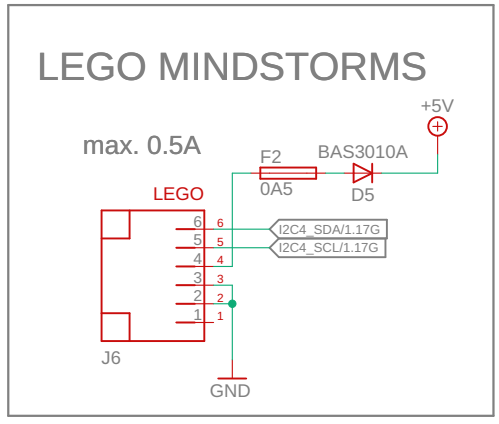
| | | | | | |
|---|---|---|---|---|---|
| Änderung | Datum | Name | Freigabe | Projekt: IR-Ring | Urspr.: Luchse-2vs2-Lightweight-Top v1.0 |
| | | | Prüfer | | Ersteller: Rafael Günther |
| | | | Datum | | Dateiname: IR-Ring |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. |
| | | | | | Version: V1.1 |
| | | | | | Speicher-Datum: 11/19/25 7:50 PM |
| | | | | | Blatt 1/6 |

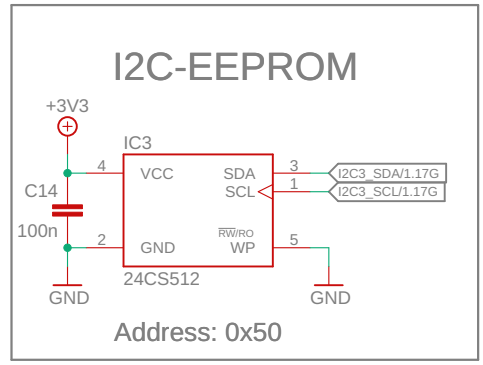
Roboter-AG BZMarkdorf

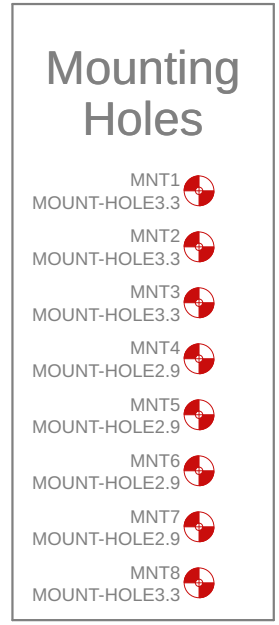


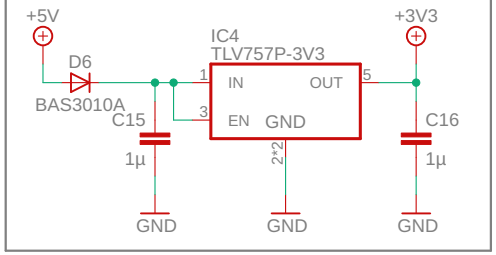




| Änderung | Datum | Name | Freigabe | Projekt: | Urspr.: Luchse-2vs2-Lightweight-Top v1.0 | Ersteller | Dateiname: |
|---|---|---|---|---|---|---|---|
| | | | Prüfer | IR-Ring | | Rafael Günther | IR-Ring |
| | | | Datum | | | Version | Speicher-Datum: |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. | V1.1 | 11/19/25 7:50 PM |

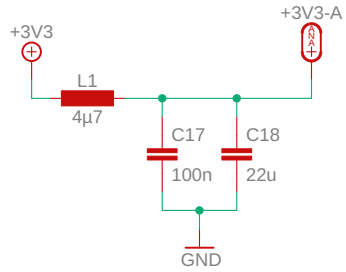
Roboter-AG BZMarkdorf

Blatt 2/6

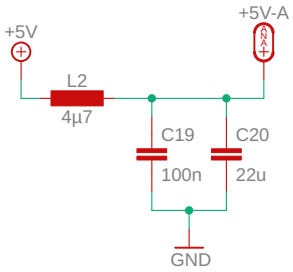
## 3V3-Supply



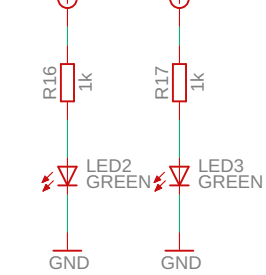

## +3V3-Analog-Supply



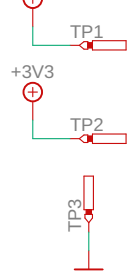

## +5V-Analog-Supply



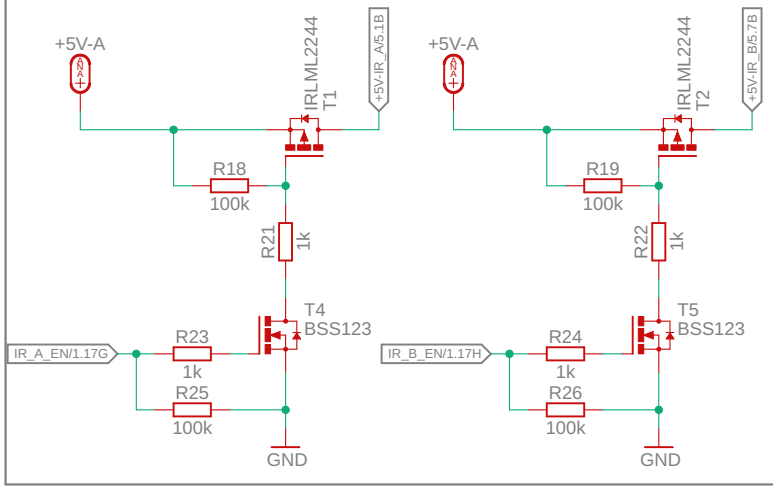

## POWER-LEDs



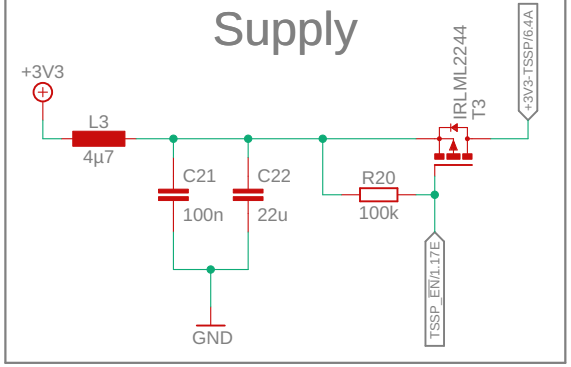

## TEST POINTS




## IR-Sensor-Supply

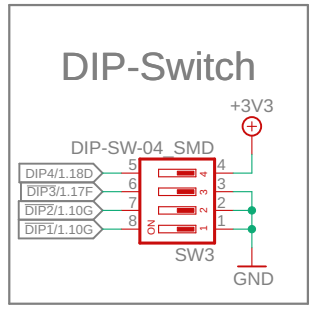


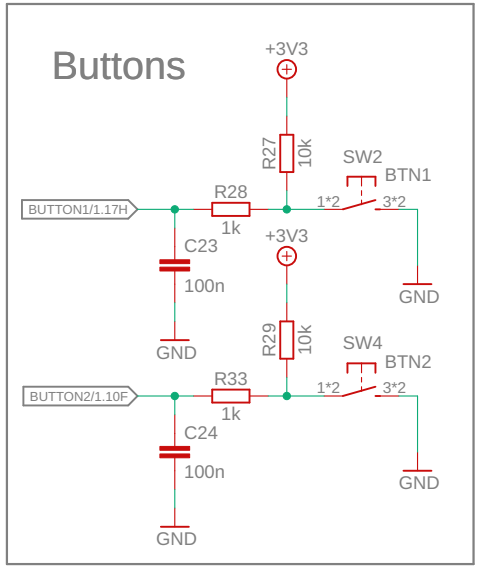

## TSSP4038-Supply

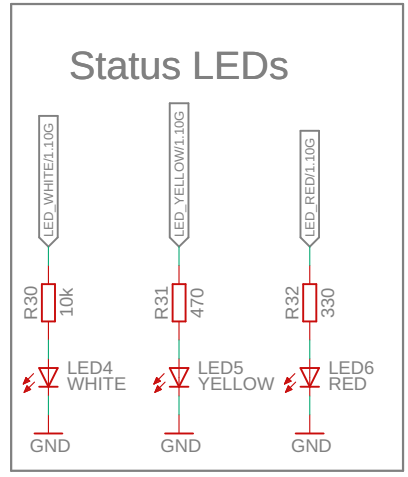


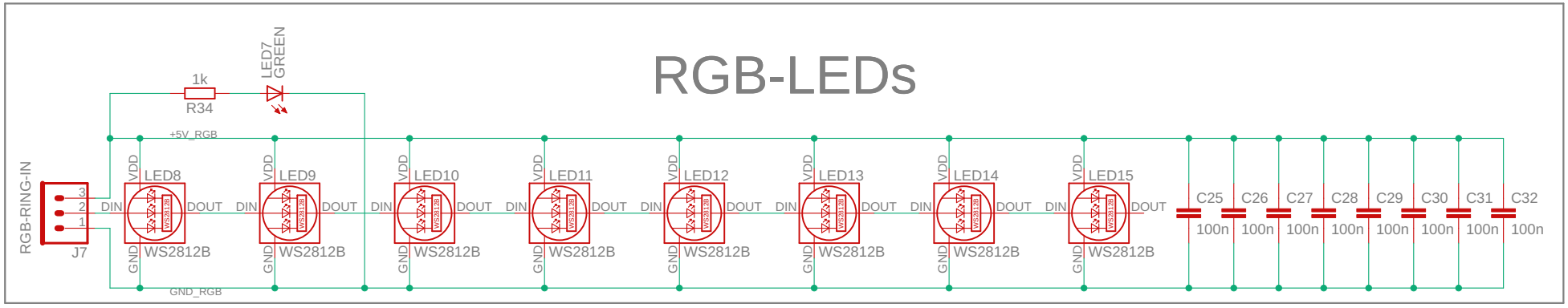

| Änderung | Datum | Name | Freigabe | Projekt: | Urspr.: Luchse-2vs2-Lightweight-Top v1.0 | Ersteller | Dateiname: |
|---|---|---|---|---|---|---|---|
| | | | Prüfer | IR-Ring | | Rafael Günther | IR-Ring |
| | | | Datum | | | Version | Speicher-Datum: / Blatt |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. | V1.1 | 11/19/25 7:50 PM / 3/6 |

Roboter-AG BZMarkdorf



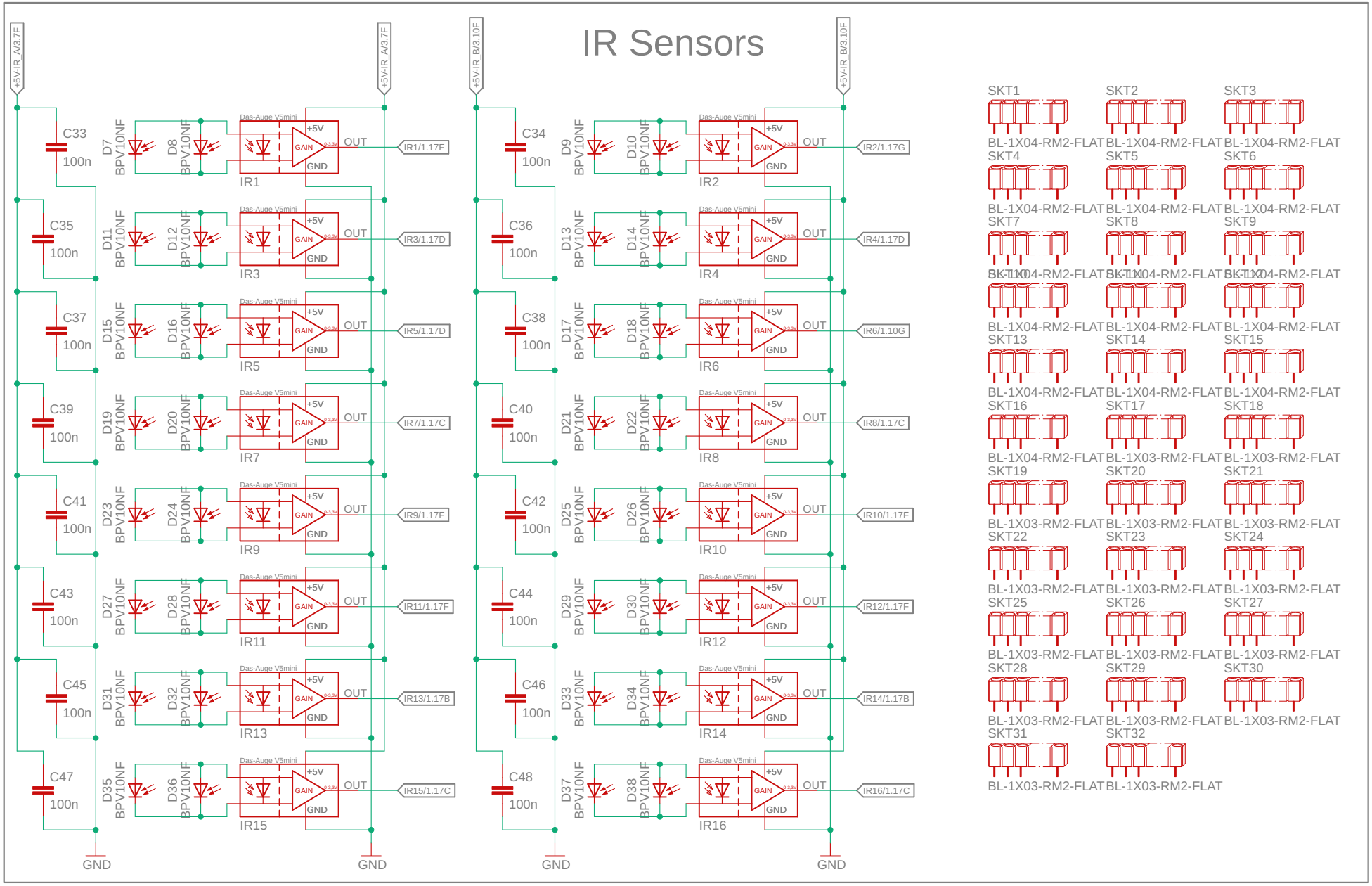




| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Änderung | Datum | Name | Freigabe | Projekt: | Urspr.: Luchse-2vs2-Lightweight-Top_v1_0 | Roboter-AG BZMarkdorf | Ersteller Rafael Günther | Dateiname: IR-Ring | |
| | | | Prüfer Datum | IR-Ring | | | Version V1.1 | Speicher-Datum: 11/19/25 7:50 PM | Blatt 4/6 |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. | | | | |

# IR Sensors







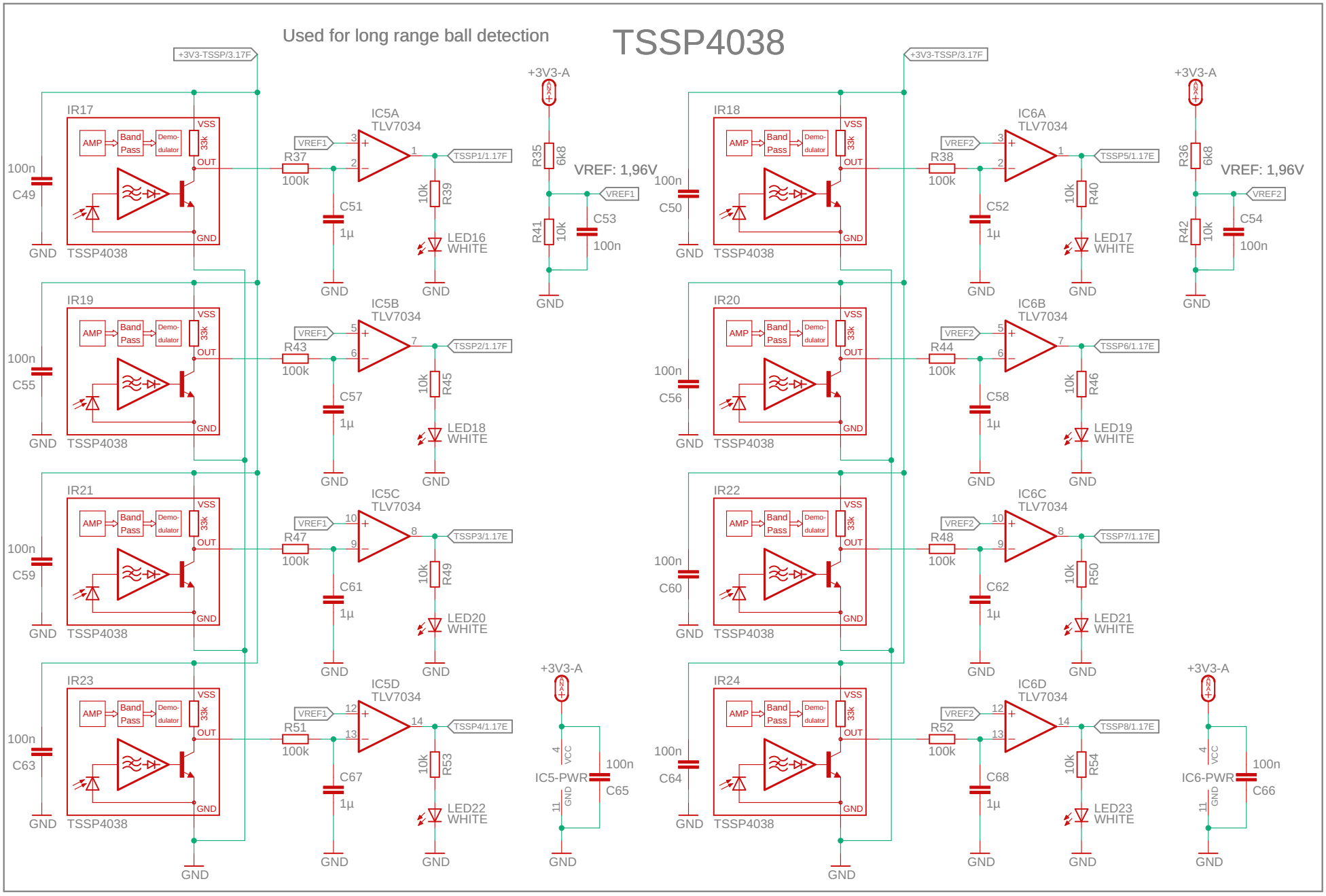
| Änderung | Datum | Name | Freigabe | Projekt: | Urspr.: Luchse-2vs2-Lightweight-Top_v1_0 | Ersteller | Dateiname: |
|---|---|---|---|---|---|---|---|
| | | | Prüfer Datum | IR-Ring | | Rafael Günther | IR-Ring |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. | Version V1.1 | Speicher-Datum: 11/19/25 7:50 PM |

Roboter-AG BZMarkdorf

Blatt 5/6

Used for long range ball detection

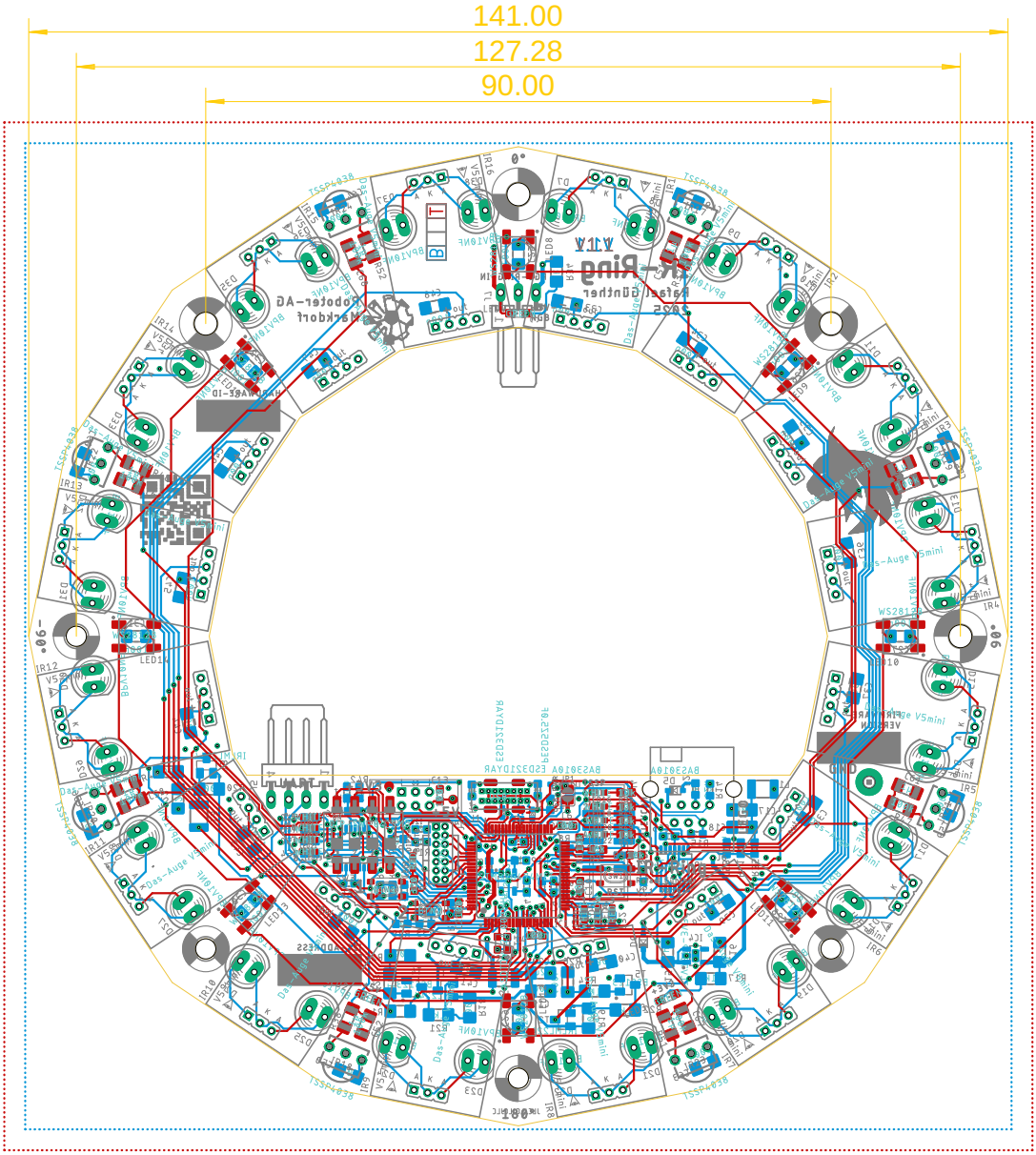
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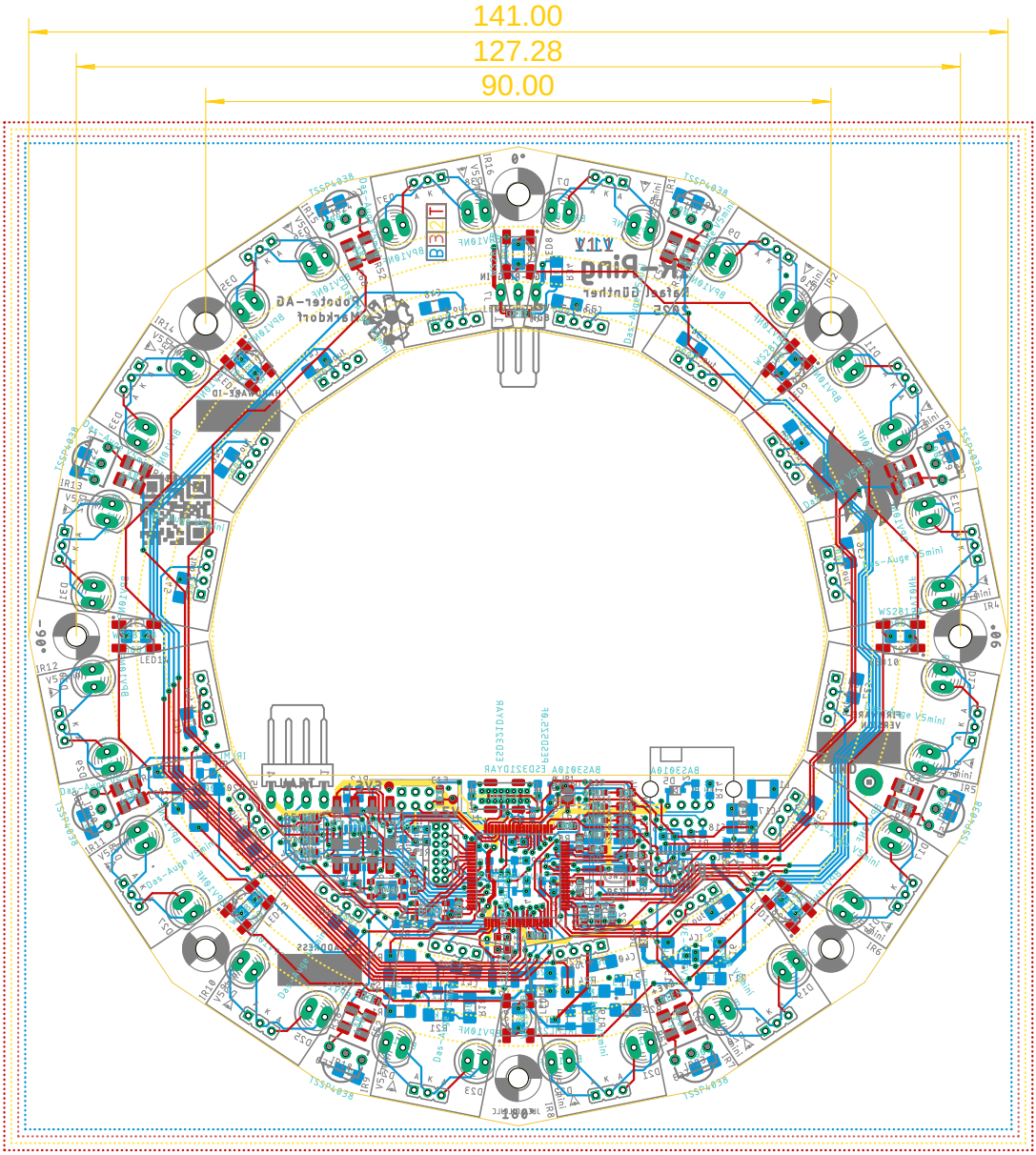


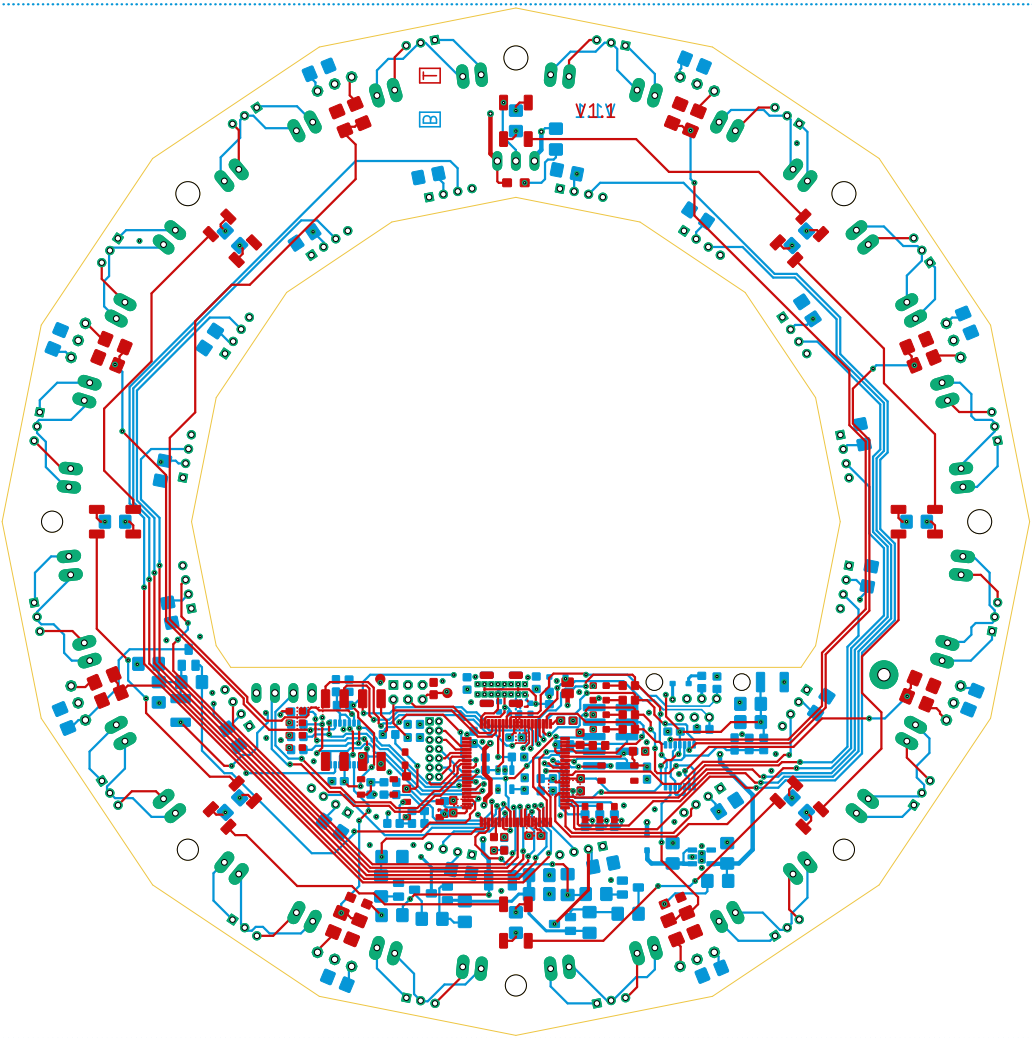

| Änderung | Datum | Name | Freigabe | Projekt: | Urspr.: | Ersteller | Dateiname: |
|---|---|---|---|---|---|---|---|
| | | | Prüfer | IR-Ring | Luchse-2vs2-Lightweight-Top v1.0 | Rafael Günther | IR-Ring |
| | | | Datum | | | Version | Speicher-Datum: |
| | | | Bearb. Rafael Günther | Ers. f. IR-Ring V1.0 | Ers. d. | V1.1 | 11/19/25 7:50 PM |

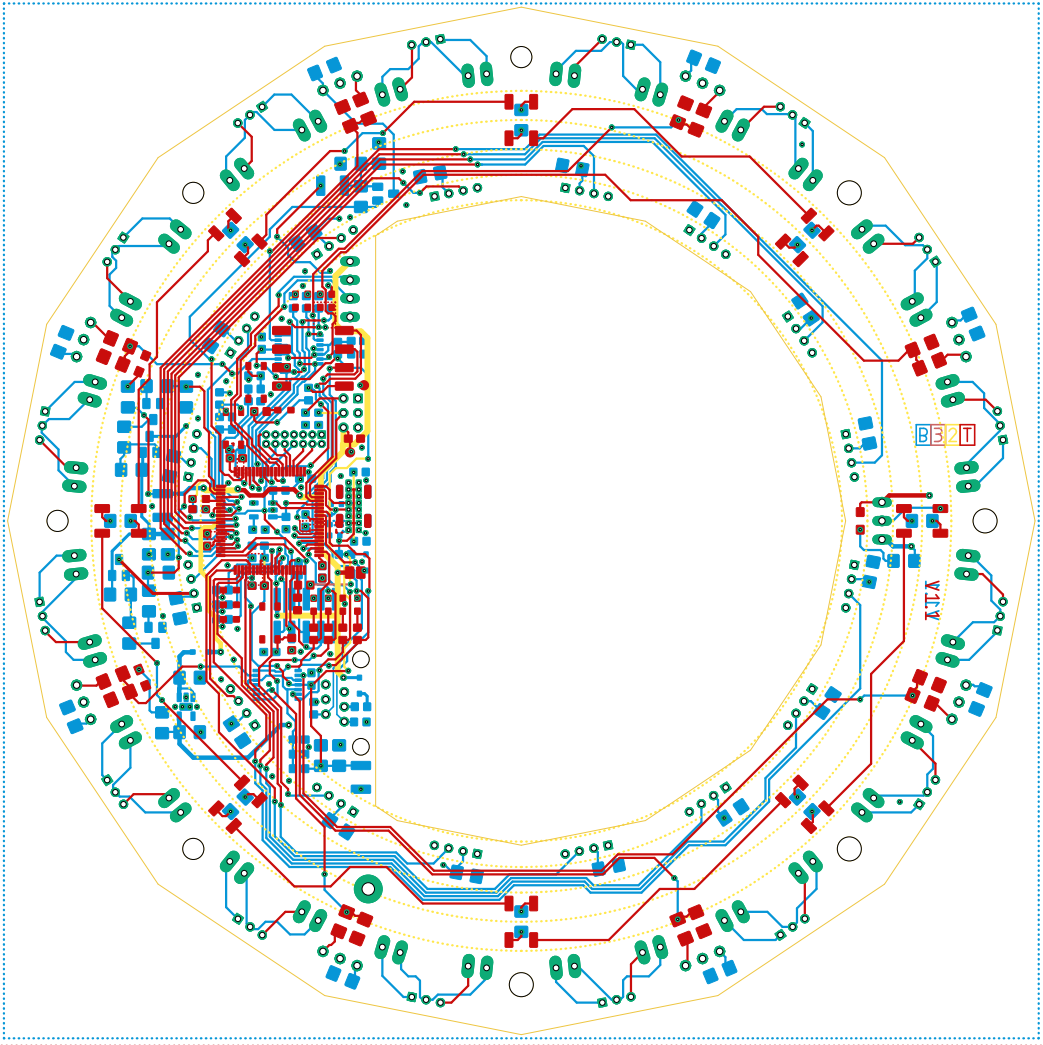
Roboter-AG BZMarkdorf

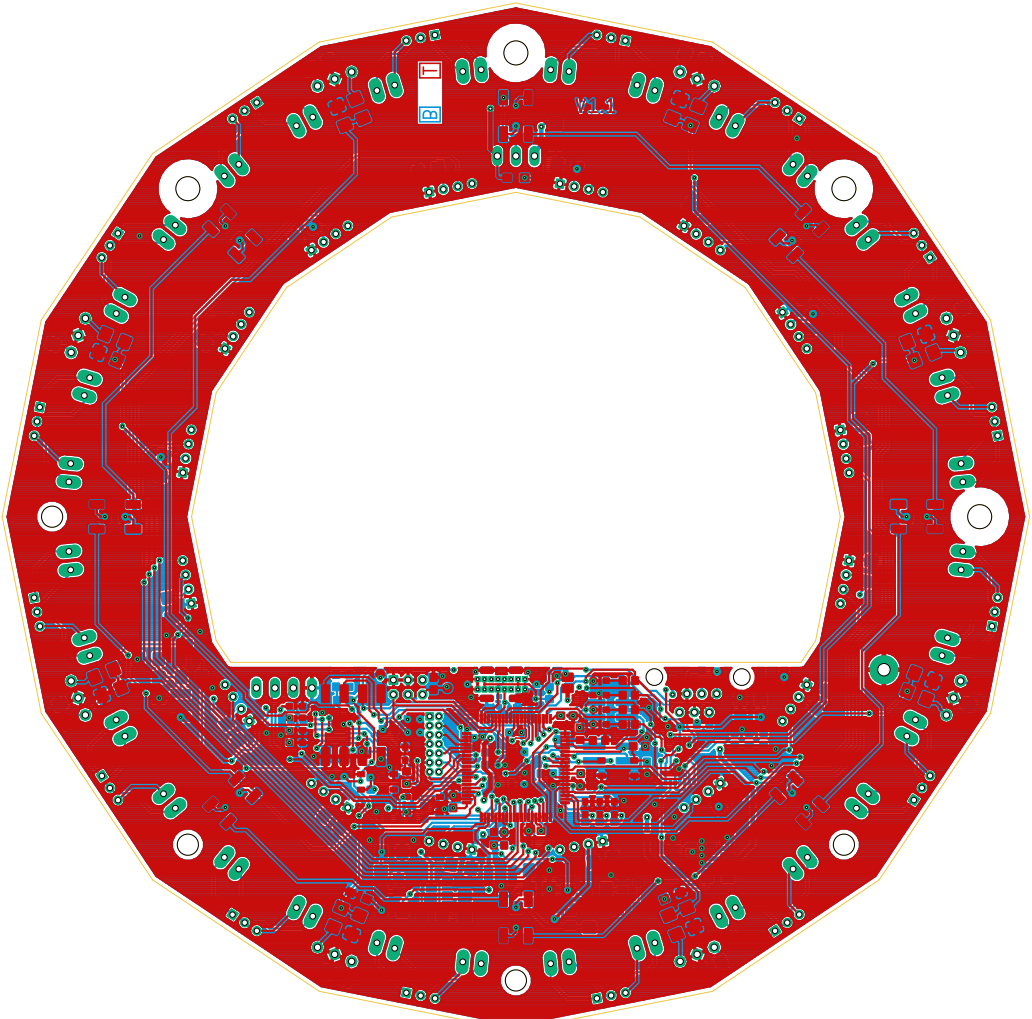
Blatt 6/6

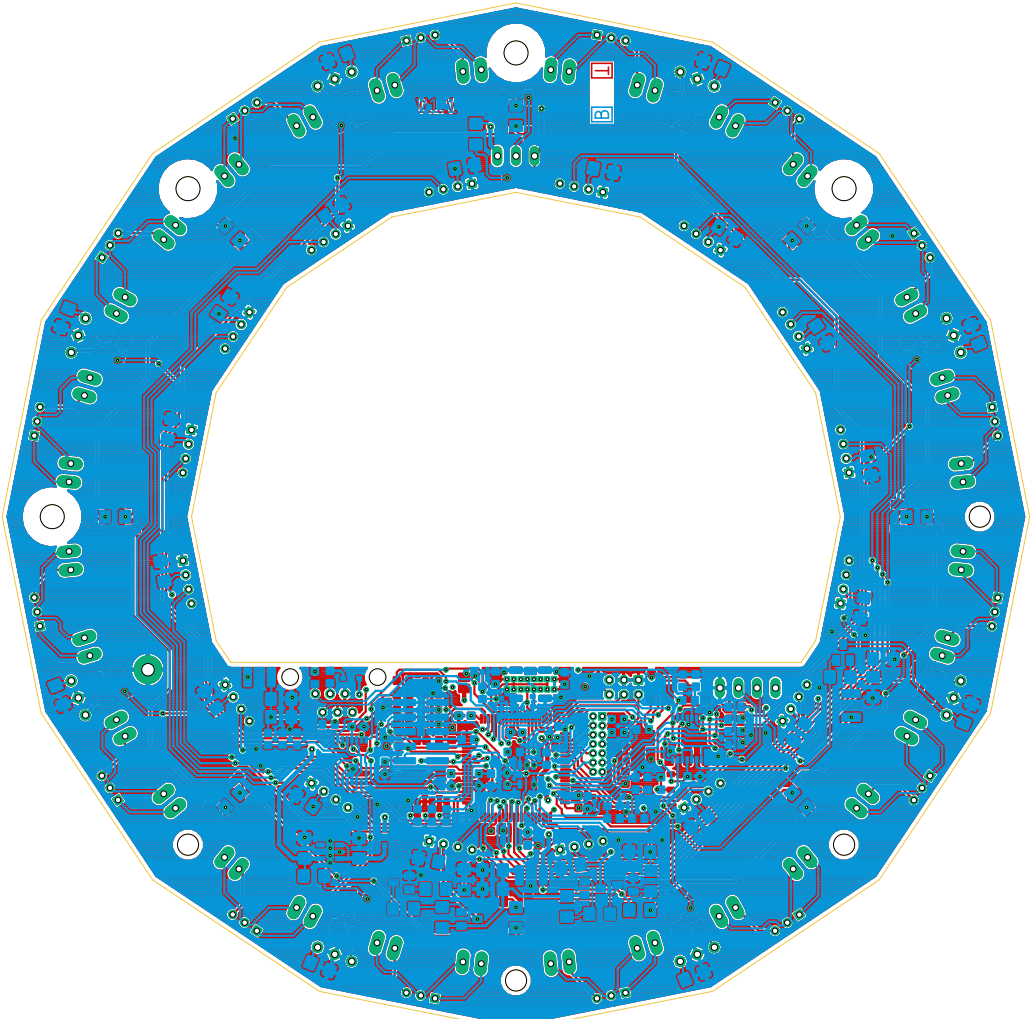

141.00
127.28
90.00


141.00
127.28
90.00




B32T
V1.1


B
T
V1.1


B1